

# Requirements Engineering

## Users

- ClubArtizen founders
- Marketing team
- anyone else?

## Stuff to ask:

1. How much data do they require us to store? All posts ever made and their metric, or just the ability to query the social media channel for the posts and info it contains?
2. Email
3. What are the competitors which we need to look up on?

## Social Media Connections Required

- Facebook/Instagram - Devshree
    - Gemini Randomly got me this link 🤖: <https://www.youtube.com/watch?v=sGOENzAOOWI>
  - LinkedIn - Co-Founders and Company Pages - Yadala
    - What exact information are we getting from the CoFounder's page? The same information we are getting from anywhere and what is its weightage?
    - Cofounder page more engagement,
    - Do we have access to all the community apis that are necessary to get all the information from LinkedIn:
      - Community Management API: <https://learn.microsoft.com/en-us/linkedin/marketing/community-management/community-management-overview?view=li-lms-2026-01>
- Information that can be gathered from LinkedIn**  
**1. Core Post Metrics (Share Statistics)**  
These are the standard "performance" metrics for every individual post:
  - **Impressions:** Total number of times the post was displayed.
  - **Unique Impressions:** The number of unique members who saw the post.
  - **Clicks:** Total clicks on the content, company name, or logo.
  - **Reactions:** Total count of all 7 LinkedIn reactions (*Like, Celebrate, Support, Funny, Love, Insightful, Curious*).
  - **Comments:** Total number of comments and replies on the post.

- **Shares:** How many times the post was reshared by others.
- **Engagement Rate:** Calculated as  $\frac{(\text{Clicks} + \text{Reactions} + \text{Comments} + \text{Shares})}{\text{Impressions}}$ .

### 3. Follower & Audience Demographics

While these are "Page Level," they tell you exactly who is seeing and engaging with your posts:

- **Geography:** Breakdown by Country and Region.
- **Company Size:** Sizes of the organizations where your audience works.

### 4. Page Traffic (Derived from Posts)

Metrics showing how posts drive traffic back to your main page:

- **Page Views:** Total views of your company page (split by Desktop/Mobile).
- **Unique Visitors:** Individual people visiting your page.
- **Button Clicks:** Clicks on your "Custom Button" (e.g., *Visit Website* or *Contact Us*).

- WhatsApp - SmartPing API - Yadala
  - Smart ping cannot be used to get any analytical information as they only provide an API key to automate the sending of certain posts but not get any analytical information as far as I could see
  - Will have to use the WhatsApp Business API and see how plausible it is get the information from it.
  - Link: <https://developers.facebook.com/documentation/links/inbound-messaging/whatsapp/overview>

### Information that can be gathered from WHATSAPP BUSINESS API

#### 1. Messaging Performance (WABA Level)

High-level volume metrics to track account health:

- **Sent / Delivered:** Volume vs. successful reach.
- **Read:** Open rates (subject to user privacy).
- **Received:** Total customer-to-business inbound volume.

#### 2. Conversation & Cost Analytics

- Will need to get access to the META apis so that we can access the necessary information soon to explore this [coordinated with Anita, waiting to get access]

- Focuses on ROI and pricing categories:
- **Categories:** Tracking by **Marketing, Utility, Authentication, or Service** (support).
  - **Direction:** Business-initiated vs. User-initiated.
  - **Cost:** Estimated spend based on these conversation buckets.

### 3. Template Performance

Specific data for pre-approved outbound messages:

- **CTR:** Clicks on "Quick Reply" or "CTA" buttons.
- **Quality Rating:** A "High/Medium/Low" score based on user feedback (blocks/reports).

### 4. Real-Time Webhooks

Granular, per-message status updates pushed to your server:

- **Status:** `sent`, `delivered`, `read`, or `failed`.
- **Error Codes:** Instant feedback on why a message didn't land (e.g., "invalid number").

### 5. Filters & Segmentation

Slicing the data for deeper insights:

- **By Geography:** Filter metrics by country.
- **By Asset:** Filter by specific phone numbers or time buckets ( `HALF_HOUR`, `DAY`, `MONTH` ).

- Youtube - Sian
- Google Ads - Sian
- Email Campaigns - need to look into - Srinivas
- E - Commerce - what exactly do they mean... their website only? - Srinivas
- Search/SEO - again what exactly? - Kushagra
- Blog Entries - what blog and what exact types of metrics? - Kushagra

## Dashboard Pages

- Statistics: Raw data
- Analytics: Insights
- Predictives

## Functional Requirements

| Function                                                                                     | Feasibility                                                                                                                                                                                           | MosCow      | Suggested Modifications/Compromises | Questions                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Utilize the provided API keys from the above Social Media to the Social Media Dashboard      | <ul style="list-style-type: none"><li>• Highly Feasibly for Social Media that have API keys, however the data that can be extracted would vary</li><li>• WhatsApp API might be a bit tricky</li></ul> | MUST HAVE   |                                     | <ul style="list-style-type: none"><li>• Does the client already have META Developer account access, and developer access necessary to acquire API keys?</li><li>• How much data do they require us to store? All posts ever made and their metric, or just the ability to query the social media channel for the posts and info it contains?</li></ul> Max 1 yr |
| A central widget showing a health bar per social media channel [Red/Yellow/Green]            |                                                                                                                                                                                                       | MUST HAVE   |                                     | <ul style="list-style-type: none"><li>• what is health?</li><li>• What exact information must this take into account?</li></ul>                                                                                                                                                                                                                                 |
| Provide a large database of usable widgets that can be dragged onto the dashboard as needed. |                                                                                                                                                                                                       | MUST HAVE   |                                     | <ul style="list-style-type: none"><li>• what exact widgets are expected?</li></ul>                                                                                                                                                                                                                                                                              |
| Ability to build Custom Widgets based on the options chosen                                  |                                                                                                                                                                                                       | SHOULD HAVE |                                     |                                                                                                                                                                                                                                                                                                                                                                 |

| Function                                                                          | Feasibility                                                                                                                                                                                                                                                                                                                                                                  | MosCow     | Suggested Modifications/Compromises | Questions                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Show heatmaps and spidermaps where ever applicable                                |                                                                                                                                                                                                                                                                                                                                                                              | MUST HAVE? |                                     | <ul style="list-style-type: none"> <li>What exact information is the client trying to gain from such graphs so that we have more information to find out what types of graphs they would benefit from.</li> </ul> <p>Able to see how effective smth is just a way to visualize all the different information, slice data by date on various parameters</p> |
| Create a database of competitors with their channels, option to add a competitor. | <ul style="list-style-type: none"> <li>Because of privacy guidelines created by social media companies, only so much information can be gathered about competitors.</li> <li>For instance, META only allows developers to access the page of any person outside the account they are connected to, and no information about clicks, views, and more sophisticated</li> </ul> | COULD HAVE | Same as below                       | <ul style="list-style-type: none"> <li>Names of Specific competitors</li> <li>What exact type of data does the client envision to compare upon.</li> </ul> <p>see what is possible exactly...</p>                                                                                                                                                          |

| Function                                                                                                                                                                                                                               | Feasibility                     | MosCow     | Suggested Modifications/Compromises                                                                                                                                 | Questions                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                        | information cannot be gathered. |            |                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                            |
| Comparisons with competitors in realtime                                                                                                                                                                                               | Same concerns as above          | COULD HAVE | Instead of realtime analysis, we could provide the trends that the competitors are focusing on, like the content they are posting - THIS IS A FINAL ADDITION THOUGH | <ul style="list-style-type: none"> <li>What do you mean by realtime specifically, in general across the whole Dashboard/product When logged in it should be latest</li> </ul>                                                                                                                                                                                                              |
| The Statistics page must contain the following for each Social Media Channel: Follower Graph, Engagement metrics [Likes, comments, shares, etc.], Impressions/Views, Click through rate (where applicable). THIS MUST BE COMPREHENSIVE |                                 | MUST HAVE  |                                                                                                                                                                     | <ul style="list-style-type: none"> <li>How comprehensive is this expected to be?               <ol style="list-style-type: none"> <li>OverAll total metrics so far, with the ability to filter?</li> <li>Is this expected of each post ever created for each social media? kind of post is doing well, analyze the post, and what is doing well... analyze the post</li> </ol> </li> </ul> |
| Paid and Organic Views                                                                                                                                                                                                                 |                                 |            |                                                                                                                                                                     | <ul style="list-style-type: none"> <li>what exactly are paid and organic views that can be seen in the META dashboard?</li> </ul>                                                                                                                                                                                                                                                          |
| Categorize top post type among: Reel, Posts, Carousel, Poll, Story (if possible)                                                                                                                                                       |                                 |            |                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                            |
| Perform Cross Correlations between different metrics like: posting time vs performance, Demographics vs                                                                                                                                |                                 |            |                                                                                                                                                                     | <ul style="list-style-type: none"> <li>How comprehensive do they expect the cross validation to be? At what level of specificity?</li> </ul>                                                                                                                                                                                                                                               |

| Function                                                                                                                                                                                                                                                                                                            | Feasibility                                      | MosCow     | Suggested Modifications/Compromises | Questions                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| performance, location vs performance etc...                                                                                                                                                                                                                                                                         |                                                  |            |                                     |                                                                                                                                                                                                    |
| Optimal posting times and optimal lengths of posts suggestions                                                                                                                                                                                                                                                      |                                                  | Could Have |                                     |                                                                                                                                                                                                    |
| Further cross and content validation: content type [reels and whatnot] vs performance, product vs performance and such...                                                                                                                                                                                           |                                                  |            |                                     |                                                                                                                                                                                                    |
| Spotlights: Products, artisans, CA spotlight                                                                                                                                                                                                                                                                        |                                                  |            |                                     | <ul style="list-style-type: none"> <li>• does the client want us to suggest post ideas that will advertise this to the outside world??</li> <li>• how exactly are we getting the inform</li> </ul> |
| <p>Create alerts for the following:</p> <ol style="list-style-type: none"> <li>1. Engagement drops 20%/ or rises 20%</li> <li>2. Competitor stats increase 25% or more (possible?)</li> <li>3. Number of unanswered comments is more than 10%</li> <li>4. No response for 24 hours for a comment from CA</li> </ol> | Feasibility of some of these must be looked into |            |                                     |                                                                                                                                                                                                    |
| <p>Create alerts for the following:</p> <ol style="list-style-type: none"> <li>1. Engagement</li> </ol>                                                                                                                                                                                                             |                                                  |            |                                     | <ul style="list-style-type: none"> <li>• where should the alerts be sent? Just a note in the dashboard or an</li> </ul>                                                                            |

| Function                                                                                                                                                                                 | Feasibility | MosCow | Suggested Modifications/Compromises | Questions                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|-------------------------------------|-------------------------------------------------------------------------------|
| drops 20% or rises 20%<br>2. Competitor stats increase 25% or more (possible?)<br>3. Number of unanswered comments is more than 10%<br>4. No response for 24 hours for a comment from CA |             |        |                                     | email or a message?<br>• need more specifics<br>• need to look more into this |
| Ability to filter data based on demographics and other information                                                                                                                       |             |        |                                     |                                                                               |
| INSIGHTS can appear against the graphic or as a list.                                                                                                                                    |             |        |                                     | • What exactly do they mean by this?                                          |

## Non Functional Requirements

### 1. Performance & Reliability

- **Data Latency:** Competitor and internal stats should refresh in "near real-time" (ideally within 5–15 minutes of the source update).
- **Availability:** The dashboard must maintain 95% uptime to ensure alerts are never missed.

### 2. Usability & UX

- **Responsiveness:** The dashboard must be fully functional on both desktop and mobile browsers.
- **Intuitiveness:** The "Two-Parameter Widget Creator" must be usable by non-technical marketing staff without coding knowledge.

### 3. Security & Privacy

- **Credential Management:** Encrypted storage of API keys and social media tokens.
- **Role-Based Access Control (RBAC):** Ability to define who can see "Co-founder" pages vs. "Company" pages.